

A scatter plot showing the relationship between GeneRatio (x-axis) and  $-\log_{10}(\text{Padj})$  (y-axis) for 1000 genes. The x-axis ranges from 0.075 to 0.175, and the y-axis ranges from 0 to 10. The plot uses a color scale to indicate the significance of the genes: blue for  $P > 0.05$ , purple for  $0.01 < P \leq 0.05$ , and red for  $P \leq 0.01$ . The data points show a clear positive correlation, with the most significant genes (red) having the highest GeneRatio and  $-\log_{10}(\text{Padj})$  values.

| GeneRatio | $-\log_{10}(\text{Padj})$ | Significance |
|-----------|---------------------------|--------------|
| 0.075     | 0.5                       | Blue         |
| 0.105     | 1.5                       | Purple       |
| 0.105     | 2.5                       | Purple       |
| 0.120     | 4.5                       | Purple       |
| 0.120     | 5.5                       | Purple       |
| 0.130     | 6.5                       | Purple       |
| 0.145     | 7.5                       | Red          |
| 0.145     | 8.5                       | Blue         |
| 0.145     | 9.5                       | Blue         |
| 0.160     | 10.5                      | Red          |
| 0.160     | 11.5                      | Red          |
| 0.160     | 12.5                      | Purple       |
| 0.160     | 13.5                      | Purple       |
| 0.180     | 14.5                      | Red          |
| 0.180     | 15.5                      | Red          |

